

ARUL S

Junior Software Developer Intern @ VRNeXGen Technologies

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SUMMARY

Information Technology graduate with hands-on experience in FastAPI, full-stack development, and machine learning projects. Skilled in building REST APIs, database-driven applications, and responsive web interfaces using modern technologies. Interested in AI Engineering, LLM applications, and scalable backend systems.

SKILLS

- **Languages:** Java, Python, JavaScript
- **Backend:** FastAPI, Node.js, Express.js, REST API, JWT Authentication
- **Frontend:** React.js, HTML5, CSS3
- **Databases:** PostgreSQL, MongoDB, Firebase Firestore, SQL
- **UI/UX Design:** Figma, Wireframing, Prototyping, User Research
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib
- **Mobile:** Android Development (Java, XML)
- **Tools:** Git, GitHub, VS Code, Android Studio, Power BI, N8N

WORK EXPERIENCE

Junior Software Developer Intern, VRNeXGen Technologies

January 2026 – Present

- Designed and built FastAPI backend APIs with a layered architecture, adding JWT-based authentication (HTTP-only cookies) and password hashing, reducing unauthorized access
- Integrated PostgreSQL via psycopg2, optimized queries, and implemented connection pooling in FastAPI, improving query response time by 25%.
- Created role-based UI/UX for 3 user types in Figma, delivering complete dashboards and workflow screens that reduced onboarding time by 40%.

EDUCATION

B.Tech – Information Technology

2022 – 2026

University College of Engineering Tindivanam | CGPA: 8.591/10

PROJECTS

Smart Contract Vulnerability Detection using Graph-Based Contract Sensing | [Github](#)

- Converted Solidity smart contracts into node-edge graphs using custom regex-based parsing for structural code analysis.
- Fine-tuned GraphCodeBERT with Masked Language Modeling (MLM) and trained a Structure-Aware Transformer-based Graph Neural Network (GNN) to detect vulnerabilities.
- Achieved 97.96% classification accuracy in automated smart contract security auditing.

Symposium Website – Dotzv12 (MERN Stack) | [Github](#)

- Built a full-stack MERN application supporting multi-event registration with a responsive UI, handling 200+ participant enrollments and payments.
- Integrated Cashfree payment gateway and Brevo email API for automated transaction handling and confirmation emails.
- Engineered end-to-end automated certificate generation and email delivery for all event participants using Brevo API, reducing post-event manual effort to zero.

Banking System (Java, JavaFX, MVC)

- Developed a banking system using Java MVC architecture with features including account creation, deposit, withdrawal, and balance inquiry.
- Built a JavaFX GUI with clear separation of business logic and presentation layer.

- Applied OOP principles — encapsulation, inheritance, polymorphism — for a modular, reusable codebase.

CERTIFICATIONS

- MERN Stack Application Development — Edunet Foundation (April 2025)
- Figma for Mobile and Web Application — Udemy (April 2025)
- Python Libraries: NumPy, Pandas, Matplotlib — Udemy (May 2025)
- MongoDB — Udemy (September 2024)
- Java — HackerRank (September 2024)
- Android App Development — Acmegrade (March 2024)

AWARDS & HONORS

- Secured Top Rank Holder in Academics – University College of Engineering, Tindivanam (2025)
- Secured First Place in Hackathon – University College of Engineering, Tindivanam (2024)

POSITIONS OF RESPONSIBILITY

- Student Placement Coordinator — University College of Engineering Tindivanam (2024 – 2026)
- Technical Symposium Coordinator — University College of Engineering Tindivanam (2025)

TRAINING & WORKSHOPS

- Workshop on Azure AI using Python — Ednue Technologies
- Workshop on Data Science — Mailam Engineering College